Through D we draw the perpendicular to AB and let it to meet the semi-circle at C. We draw straight lines CA, CB and CM. Since M is the center $CM =$ radius $= \frac{a+b}{2}$. Clearly $MD = \frac{a+b}{2} - a$. Using the similar triangles ΔACD and ΔCBD we have $\frac{CD}{AD} = \frac{BD}{CD}$ and hence $CD^2 = AD \times BD = ab$. So $CD = \sqrt{ab}$. (Using Pythagoras theorem also we can prove that $CD = \sqrt{ab}$.) Since the length of any half chord is less than or equal to the radius, we have $CD \leq CM$. In other words, $\sqrt{ab} \leq \frac{a+b}{2}$. This means $GM \leq AM$.

The length of the half chord DC is equal to the radius if and only if $D = M$. Thus equality $AM = GM$ holds if and only if $a = b$.

Result 5.1: If $a_1, a_2, a_3, \ldots, a_n$ is an arithmetic progression, every term $a_k \ \ (k > 1)$ is the arithmetic mean of its immediate predecessor a_{k-1} and immediate successor a_{k+1}.

Proof. Let $a_1, a_2, a_3, \ldots, a_n$ be an arithmetic progression with initial term a and common difference d. Then

$$a_k = a + (k-1)d, \quad a_{k-1} = a + (k-2)d \ \text{ and } \ a_{k+1} = a + kd.$$

Thus

$$\frac{a_{k-1} + a_{k+1}}{2} = \frac{a + (k-2)d + a + kd}{2} = \frac{2a + (2k-2)d}{2} = a + (k-1)d = a_k.$$

Therefore, a_k is the arithmetic mean of a_{k-1} and a_{k+1}. ■

Result 5.2: If $a_1, a_2, a_3, \ldots, a_n$ is a geometric progression, every term $a_k \ \ (k > 1)$ is the geometric mean of its immediate predecessor a_{k-1} and immediate successor a_{k+1}.

Proof. Let $a_1, a_2, a_3, \ldots, a_n$ be a geometric progression with initial term a and common ratio r. Then

$$a_k = ar^{k-1}, \quad a_{k-1} = ar^{k-2} \ \text{ and } \ a_{k+1} = ar^k.$$

Thus

$$\sqrt{a_{k-1}\, a_{k+1}} = \sqrt{ar^{k-2} ar^k} = \sqrt{a^2 r^{2k-2}} = ar^{k-1} = a_k.$$

Therefore, a_k is the geometric mean of a_{k-1} and a_{k+1}. ■

Example 5.14 Find seven numbers $A_1, A_2, \ldots, A_7$ so that the sequence $4, A_1, A_2, \ldots, A_7, 7$ is in arithmetic progression and also 4 numbers G_1, G_2, G_3, G_4 so that the sequence $12, G_1, G_2, G_3, G_4, \frac{3}{8}$ is in geometric progression.

Solution:
Since $a = 4$ and $4 + 8d = 7$ we get $d = \frac{3}{8}$. So the required 7 numbers are $4\frac{3}{8},\ 4\frac{6}{8},\ 5\frac{1}{8},\ 5\frac{4}{8},\ 5\frac{7}{8},\ 6\frac{2}{8},\ 6\frac{5}{8}$.

Since $a = 12$ and $ar^5 = \frac{3}{8}$ we get $r^5 = \frac{1}{32}$ and hence $r = \frac{1}{2}$. Thus the required 4 numbers are $6, 3, 1\frac{1}{2}, \frac{3}{4}$.

Example 5.15 If the product of the 4th, 5th and 6th terms of a geometric progression is 4096 and if the product of the 5th, 6th and 7th-terms of it is 32768, find the sum of first 8 terms of the geometric progression.

Solution:
Let $a, ar, ar^2, \ldots$ be the geometric series having the given properties. Since the 4th, 5th and 6th terms are ar^3, ar^4 and ar^5, their product is a^3r^{12}. Thus $a^3r^{12} = 4096$. Similarly $a^3r^{15} = 32768$.